

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

November 2017

EE405 Power System Protection

Date: 29/11/2017, Wednesday

Time: 01.30 p.m. to 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Draw the block diagram of carrier blocking scheme used with distance relay for transmission line protection and explain it. [07]

Q – 2. Attempt any TWO

A The part of the power system is shown in figure 1. Decide the three zone settings of a mho distance relay R having characteristic angle of 65° . First zone covers 80% of transmission line L_2 , second zone reach is up to bus C and third zone covers 50% of longest transmission line after bus C. [07]

Required data is as follows:

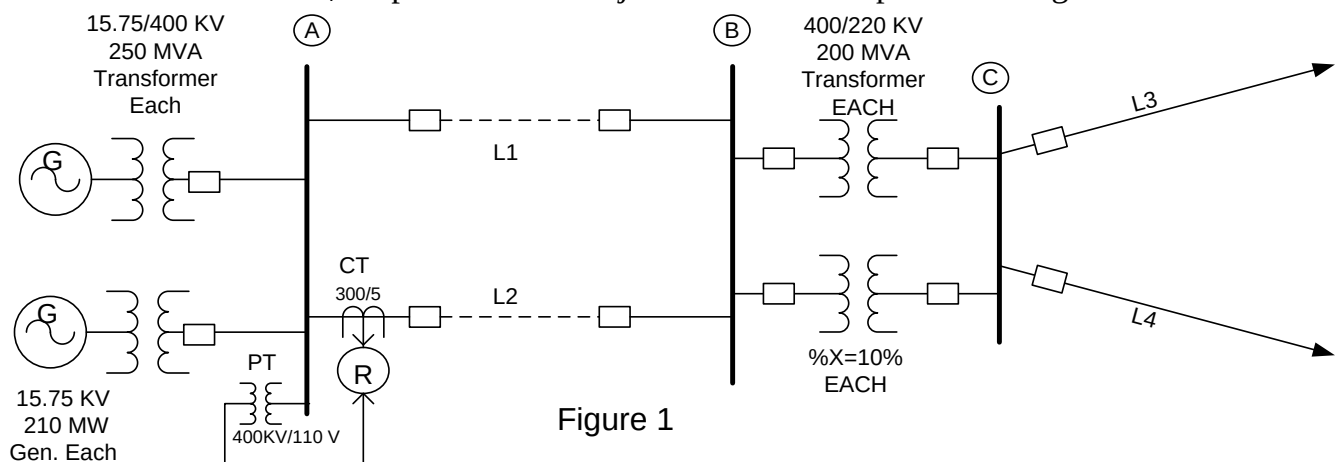
(i) 400 KV Lines L_1 and L_2

Impedance: $0.02 + j 0.1 \text{ ohm / km / ph.}$ Line length: 300 Km.

(ii) 220 KV lines L_3 :

L_3 : Impedance: $0.04 + j 0.12 \text{ ohm / Km / ph.}$ Line length: 150 Km

L_4 : Impedance: $0.06 + j 0.15 \text{ ohm / Km / ph.}$ Line length: 125 Km



B Draw neat and well labeled diagram for duplicate busbar protection scheme. In this diagram trace the fault current through CT secondary circuit for any in-zone fault. [07]

C Explain with mathematical proof: How the phasors are estimated for relaying quantity. [07]

Q – 3 Attempt any TWO

A Following are the details of 3 phase, 800 HP, 6 KV Induction motor. **[07]**

- i. Rated p.f.:0.85
- ii. Efficiency: 80%
- iii. Starting time: 6 seconds
- iv. Starting Current: 5.5 times the rated current
- v. CT ratio: 100/1 Amp.
- vi. Motor can able to take 110% of rated current continuously.

Relay: Rating =1 amp.

- a) Thermal O/L Unit: 70%-130% of 1 Amp in step of 10%
- b) Instantaneous O/C Unit: 400%-1500% of 1 Amp in step of 50%
- c) NPS Unit: 100% -600% of 1 Amp in steps of 50%
- d) Stalling Unit:
 Current Setting Range: 90-600% of 1 Amp in step of 30%
 Time Setting Range: 4-30 seconds.
 (Safe stalling time of I.M. Hot=13 sec; Cold=21 Sec.)
- e) Under Voltage Relay: Range 70-100% of 110 V. (P.T. 6 KV/110V)

Suggest the setting of above mentioned relay for proper protection of I.M.

B A 15.75 KV, 3 phase, 50 Hz, 210 MVA, star connected generator is protected by the simple differential protection. The CTs used are 7500/5 A. The relay is set to operate for a current of 0.15 A. **[07]**

Through fault current is 10 times the rated current. Under through fault condition, the CT₂ secondary voltage is 80% of CT₁ secondary voltage.

The relay having a resistive impedance of 100 ohms is connected to 100 meters from CT₁. The lead wire has a resistance of 0.5 ohm per 100 meters. The distance between the two sets of CTs is 250 meters.

Determine the stabilizing resistance required to be connected in series with relay to have a stability factor of 3 for this fault condition.

C Give Reason:

- i. Magnetising inrush current of a transformer creates a problem in differential protection of a transformer. **[02]**
- ii. A short-circuited lag coil is wound on one of the outer limbs of E-core of induction disc type over-current relays. **[02]**
- iii. Two overcurrent and one earth fault scheme of protection does not provide adequate protection for a transformer feeder. **[03]**

SECTION – II

Q – 4. With neat and clean diagram, show that how distance relay can be saved from mal-operation during blowing of fuse in the PT circuit. **[07]**

- Q – 5. A** Figure 2 shows a single line diagram of a part of power system. Relay used are IDMT over-current relays and rating of relays are as per CT secondary rating. Setting range of relays are 50-200% of rated current in step of 25%. Determine the plug setting of all relays and TMS of R_1 , R_2 and R_3 if TMS of R_4 and R_5 both are set to 0.1 [08]

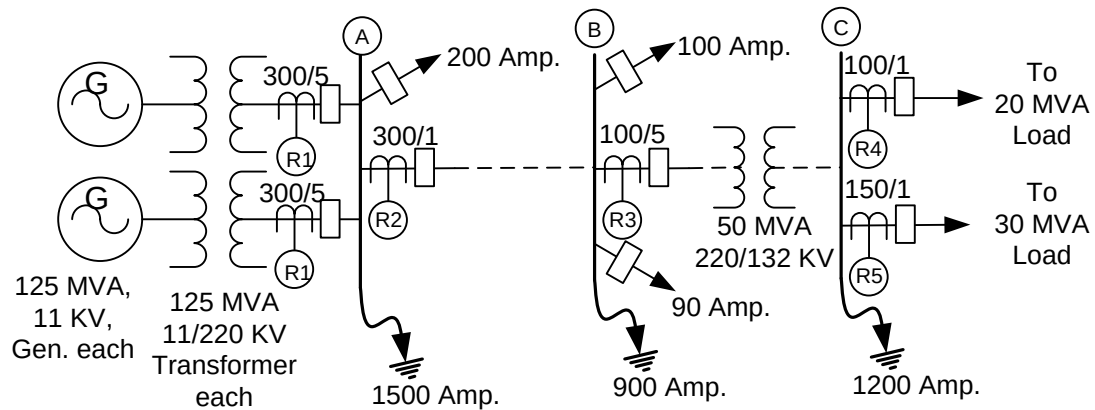


Figure 2

B Give Reason

[06]

- Directional overcurrent relay will not operate even for a fault in operative direction if there is a bolted 3-phase short circuit fault at the breaker terminals.
- In case of applying directional O/C protection, generally 90° connection of directional unit is preferred.

OR

- Q – 5. A** Figure 3 shows a single-line diagram of a power system. Bracketed values give the fault levels on the concerned bus. Relays used are IDMT over current relays. Relays are rated as per CT secondary with a setting range of 50-200% of relay rating in 7 equal steps each of 25%. Relay R_4 is set at a TMS of 0.20. Assuming a suitable discrimination time, determine the plug-settings and time-settings of all the relays shown in Figure. [08]

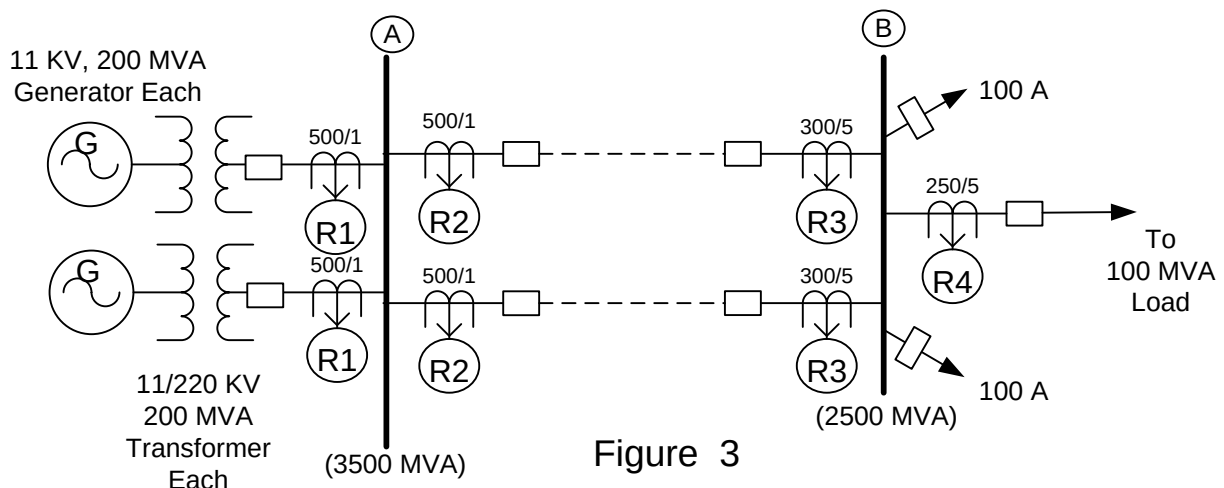


Figure 3

B Give Reason [06]

- (i) Only current discrimination is not sufficient to protect the transmission lines.
- (ii) Instantaneous overcurrent relays cannot be applied for very short line sections.

Q – 6. A A transformer is protected by biased differential relay. The relevant details are as per following. [07]

i. Transformer:

310 MVA, 220/400 KV, 3 phase, 50 Hz, DY_{11} , Neutral solidly grounded.

Maximum through fault current is 14 times the rated current.

$I_{mag\ Inrush} = 10 \times I_{Rated}$, $I_0 = 3\%$ of I_{Rated} .

On load Tap changer = -10% to +7.5% in step of 5% ,

ii. Current transformers:

L.V. Side: 1000/1A Star connected, H.V. Side: 500/1 A Star connected.

CTs are of class 3

iii. Biased Differential Relay: (Relay rating 1 Amp.)

Basic setting: 5-25% of 1 amp in step of 5%

Bias setting: 10-50% in step of 10%

H.S. inst. setting 800 to 1600% of 1 Amp in step of 100%

Find out the ICT ratio and draw the detailed 3- Φ diagram of protective scheme. Also suggest the relay setting.

B List out and explain the factors to be considered while selecting the CT for power system protection purpose. [07]

OR

Q – 6. A Why the ground fault near to the neutral of the generator must be sensed by protection scheme? Draw and explain the 100% stator earth-fault scheme for protection of large generators. [07]

B Enlist the special tests that are performed on the protective relays. Explain each of them. [07]
